

ENCN423/ENCN627 - Sustainable Energy Systems

Rebecca Peer & Rasul Satymov

Module 10: Fuels

Summary

This module explores hydrogen as a versatile, promising energy carrier and feedstock. From fertilisers to energy storage systems, hydrogen has diverse applications in the transition towards a more sustainable energy future. However, not all hydrogen is created equally, and it is mostly produced from fossil fuels today. Through this module, students will gain an understanding of hydrogen's properties, production methods, hydrogen derived fuels, and the feasibility of their potential applications, as well as the challenges and opportunities of integrating into our energy systems.

Objectives

After this module, you should:

1. Understand the potential role that renewable fuels can have in energy systems.
2. Understand the fundamentals of hydrogen production via electrolysis.
3. Be able to calculate the levelised cost of hydrogen (LCOH₂).
4. Understand the operational challenges of generating, transporting and using e-fuels.
5. Be able to communicate arguments in support and against different renewable fuel applications.

Literature

Recommended:

- Handbook of Energy Storage, Chapter 8.0 ([link](#)).
- Green Hydrogen for Industry: A Guide to Policy Making. (IRENA) ([link](#)).
- Cleaning Up (2022). Audioblog Episode 8 “The unbearable lightness of hydrogen” ([link](#)).

References:

- Sustainable Energy: Choosing Among Options, Chapter 8.3.5 and 16.3 ([link](#)).
- The Future of Hydrogen (2019). IEA report ([link](#)).
- The Clean Hydrogen Ladder (2021) ([link](#)).

Key Points

Hydrogen is a versatile element with many potential applications as a clean and sustainable vector of energy. As the world focuses more on reducing carbon emissions and transitioning to a low-carbon economy, the role of hydrogen is becoming increasingly important. However, hydrogen is not found in a free state on Earth, and must be produced from other compounds through various processes.

Green hydrogen, produced through the process of splitting water using renewable electricity, is a particularly promising option. Despite its potential, not all hydrogen applications are feasible or competitive at this time. Furthermore, hydrogen is often not the end product, but rather used to produce other things. This is why the concept of “Power-to-X” is often used, meaning that electricity (Power) is used to make other stuff (X). In the following paragraphs, we will explore the fundamentals of hydrogen, its various production methods, and its potential applications.

Basics

Hydrogen is the most abundant element in the universe and is typically found in molecular form, H_2 . It is a colorless, odorless, and tasteless gas that is highly flammable and can burn with an invisible flame. Hydrogen has the highest energy content per unit mass of any liquid fuel (gravimetric density), and its combustion produces only water vapour as a by-product, making it a clean combustion. However, because hydrogen is not found in a free state on Earth, it must be produced from other compounds, such as natural gas, coal, or water, through various processes like steam methane reforming, coal gasification, or electrolysis. This production process can have environmental impacts depending on the source of the feedstock and the energy source used to produce hydrogen.

Green hydrogen is produced through the process of electrolysis, where an electric current is used to split water into hydrogen and oxygen. The electricity used for this process is generated from renewable energy sources such as wind power or solar PV, which makes green hydrogen a clean and sustainable source of energy. While there are many potential applications for hydrogen, not all of them are feasible or competitive at this time. It is important to carefully evaluate the feasibility and competitiveness of various hydrogen applications based on factors such as technology readiness, cost, efficiency, safety, and environmental impact. The cost of producing green hydrogen is still relatively high compared to grey or black hydrogen (from natural gas and coal, respectively), but current research and aggressive deployment plans are expected to bring down green hydrogen costs to the level of grey hydrogen costs within a couple of years.

Although it is commonly used as an umbrella term, there are many things that fall under the concept of “Hydrogen”, such as technologies that create, store, and transmit it. Additionally, there are technologies that reconvert it back to electricity or other fuels

or chemicals. This is why the concept of “Power-to-X” should be preferred, meaning that electricity is used to make a wide range of other products. Hydrogen is often not the end product, but rather used to produce other things, such as ammonia (fertilisers), or e-fuels for aviation.

Motivation

Hydrogen is receiving significant attention worldwide, with many countries focusing on incorporating it into their energy systems. While the cost of hydrogen production is decreasing quickly, the main challenge lies in transporting it over long distances. This is due to its low volumetric energy density, which refers to the amount of energy per unit volume (not to be confused with its high gravimetric energy density, which refers to the amount of energy per unit weight).

Despite these challenges, hydrogen is often considered crucial for achieving 100% renewable energy systems, especially for “unavoidable” applications where alternatives are limited or non-existent.

When it comes to New Zealand and the potential role of hydrogen, several factors make the country a promising candidate for production:

- a) ample land and low population density (without a significant land constraint)
- b) abundant renewable energy resources (such as solar, hydro, wind, geothermal, and biomass)
- c) high shares of RE in New Zealand’s power sector – on track to achieve a fully renewable power sector within five years.

This means that any additional unit of energy put into e.g. hydrogen will also be renewable (this argument does not hold as easily in countries with lower shares of renewables).

Not every country has access to such favourable conditions. Many nations are still far from achieving fully renewable power systems, which makes producing renewable hydrogen more difficult. This puts New Zealand in a favourable position to potentially become a hydrogen (or hydrogen-derivatives) exporter, providing renewable hydrogen to countries that lack the same resources, infrastructure, and space. Although New Zealand’s remote location presents challenges for shipping pure hydrogen, shipping hydrogen-derived fuels can be a feasible solution, and is currently being pursued by industry.

History

In 1800, it was discovered that applying electric current (using Voltaic pile) to water could produce hydrogen and oxygen gases through a process called electrolysis, an experiment some of you might remember from high school. Reverting the process is the fuel cell effect, which combines hydrogen and oxygen gases to produce water and an electric current. This was discovered in 1839. Hydrogen has been used for various purposes throughout history, including in trans-oceanic dirigible flights (Hindenburg) and rocket propulsion, and even to generate on-board electricity, heat, and water for astronauts. Currently, the largest demands for hydrogen are oil refining, ammonia, methanol, and steel.

Despite the fuel cell technology being almost 200 years old, it is only now taking off due to improvements in electrolyzers and lower energy costs. The cost of producing hydrogen mainly involves a) capital expenditures (CAPEX) of the electrolysis technologies and b) the cost of electricity, both of which are falling rapidly. In addition, there are many incentive programs trying to push the costs of green hydrogen below the level of fossil hydrogen.

Applications

There are many applications for hydrogen. Some applications are a good idea, some are not.

The BNEF analyst Liebreich has most famously ranked the hydrogen applications in the “green hydrogen ladder”, which is under continuous updating with the quickly evolving research surrounding the topic (v5 as of May 2026). The ladder ranges from unavoidable applications (top rang) – because there are no or only little real alternatives for decarbonization- to uncompetitive (bottom rang) – because there are better or cheaper alternatives in place.

Important examples of unavoidable applications include fertilizers and methanol (which have been in high demand for decades but are currently being produced from grey or black hydrogen), as well as long-term energy storage (especially in those countries without significant hydropower reservoirs), and e-fuels for those hard-to-electrify applications (like long-haul aviation or container ships). Examples of uncompetitive applications are fuel cell electric vehicles because battery electric vehicles are superior (in terms of efficiency, infrastructure, scale, safety, and others). A similar argument can be made for heating houses with hydrogen, because electric heat pumps are far superior.

Especially in the early years of the energy transition, prioritising hydrogen applications is crucial. Adding to the green hydrogen demand implies requiring more renewables to the grid. Perhaps down the road (in the far future), in a system with energy abundance or “too cheap to meter”, we can be more generous with the hydrogen applications.

Fundamentals of production

Hydrogen can be produced from water through the process of electrolysis using two main types of electrolyzers: PEM (proton exchange membrane) and AEL (alkaline electrolysis), with the latter being the incumbent technology.

- PEM electrolyzers utilize a proton exchange membrane as the electrolyte, enabling efficient hydrogen production due to its high ionic conductivity. While the platinum-based electrodes employed in PEM electrolyzers are costly, they boast a long lifespan. Notably, from the perspective of renewable energy systems, PEMs offer quick ramping up and down.
- AEL electrolyzers use an alkaline electrolyte solution and less expensive materials for the electrodes, but they have lower efficiency and require more maintenance.

While there are other ways of producing green hydrogen, they are on a lower technological readiness level.

Hydrogen can be further converted. For example:

- Converted back to electricity via fuel cells (like an electrolyser in reverse) or via combustion in modified gas turbines
- Combined with a carbon to create methane (methanation)
- Further combined with more carbons to create e-fuels (via Fischer-Tropsch)
- Combined with nitrogen to create ammonia (via Haber-Bosch)

All the above processes are covered in detail in the “Handbook of Energy Storage”.

Economics

The cost of hydrogen production is an important factor to consider when evaluating its economic feasibility. An important metric for comparing different production methods is the levelised cost of hydrogen (LCOH₂) described in equation 1. The LCOH₂ is the total cost of hydrogen production discounted to present, divided by the annual amount of hydrogen produced.

$$LCOH_2 = \frac{CAPEX \cdot crf + OPEX + LCOE \cdot Input_{el}}{Output_{H_2}} \quad (1)$$

Where:

$CAPEX$	=	capital expenditure (\$/kW)
crf	=	capital recovery factor
$OPEX$	=	fixed operational expenses
$LCOE$	=	levelised cost of electricity
$Input_{el}$	=	electricity consumption
$Output_{H_2}$	=	annual hydrogen output

$$crf = \frac{WACC \cdot (1 + WACC)^{Lifetime}}{(1 + WACC)^{Lifetime} - 1} \quad (2)$$

Where:

$WACC$ = weighted average cost of capital (discount rate)

When producing hydrogen via electrolysis, the cost of the electrolyzers and electricity are the main drivers of LCOH₂.

The cost of electrolyzers is expected to decrease over time due to technological improvements and economies of scale, following a learning curve.

The cost of electricity is also a critical factor in determining the economic viability of hydrogen production. With the increasing deployment of renewable energy and the resulting reduction in electricity prices, the cost of hydrogen produced via electrolysis is becoming more competitive.

To optimize the economic performance of a hydrogen production plant, it is important to choose the appropriate operating strategy. When producing hydrogen via electrolysis, there is a trade-off between maximizing the use of your infrastructure and minimizing your operating costs.

On one hand, you can run your electrolyser for the maximum number of hours in a year (8760 hours) yielding more hydrogen produced. However, this also means that you have to purchase electricity during all 8760 hours, which may result in buying electricity at high prices during peak demand periods.

On the other hand, you can operate your electrolyser exclusively during times when electricity prices are low, which would result in lower operating costs. However, this also means that you would not be fully utilizing your infrastructure and would therefore produce less hydrogen overall.

Finding the optimal operating strategy that minimizes LCOH_2 can be supported by using mathematical models and simulations that take into account the variability of electricity prices and other factors such as maintenance costs and capacity utilization. By doing so, the economic feasibility of hydrogen production can be enhanced. These strategies have the co-benefit of helping the integration of renewable energy, because the electrolyzers would turn off in periods of peak demand and operate at full capacity when there would be curtailment of electricity otherwise. This is comparable to other strategies of demand-side management.

Exporting hydrogen?

The low volumetric energy density of hydrogen makes transportation difficult and expensive. There are five alternatives for transporting hydrogen over long distances:

- Hydrogen as a gas on ships. “Suiso Frontier”, the largest hydrogen tanker right now, holds about 0.2% of a LNG tanker (which to be fair are much larger in volume).
- Hydrogen as a liquid. The liquid hydrogen tanker, from Kawasaki, holds about 4x less energy than a LNG tanker. Besides the volume issue, liquefying hydrogen (-253°C) requires lots of energy.
- Hydrogen as a derivative fuel. Converting hydrogen to ammonia or methanol will make it easier for transportation, but has significant losses inherent to each conversion. Nonetheless, exporting hydrogen as ammonia or methanol can make a lot of economic sense if your end-demand is ammonia or methanol. This is currently being pursued by the industry.
- Hydrogen as a gas in pipelines. The current gas infrastructure can be used with some adaptations. The major issue is that it needs to operate at lower pressures, meaning a fraction of the transported energy. Still, this is the most promising option when distances are small enough to justify a pipeline. Europe’s main hydrogen transportation projects are currently betting on pipelines.
- Hydrogen in an end-product. Hydrogen will likely be produced in significant quantities, in those countries with future cheap electricity. It might make sense to produce

the final end-product right there, especially if it is an energy-intensive industry like the case of steel.

Outlook

In the near future, the cost of producing green hydrogen is likely to become comparable to that of grey hydrogen (without carbon capture) and blue hydrogen (with carbon capture). To reach parity, numerous subsidy programs have been announced and implemented by countries worldwide. These aim to push hydrogen projects along the learning curve.

Contrary to previous periods of hypes of hydrogen, the current wave is different because of cheap electricity, cheap electrolyzers, the pressing need to defossilise, and since the war in Ukraine and Iran: energy security.

Stoichiometry

Have you ever wondered how much water and other resources are required to produce hydrogen? In this section, we'll take a closer look at the stoichiometry of hydrogen production, which is the study of the amounts of reactants and products involved in chemical reactions. We'll explore the water requirements for different hydrogen production methods (steam methane reforming and electrolysis), as well as the amount of hydrogen needed to produce other chemicals like methanol and ammonia. By understanding the stoichiometry of these processes, we can gain a better appreciation for the resources required to produce the fuels we rely on.

How much water is needed to produce hydrogen?

1. **Steam methane reforming:** requires 4.5 L of water per kg of hydrogen

- Steam reforming: $CH_4 + H_2O \rightarrow CO + 3H_2$
- Water gas shift: $CO + H_2O \rightarrow CO_2 + H_2$
- Overall: $CH_4 + 2H_2O \rightarrow CO_2 + 4H_2$

Note that this is the theoretical stoichiometric mass consumed in the chemical reaction. The process of steam reforming will also require significant water for steam production and system cooling.

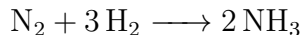
2. **Electrolysis:** Producing hydrogen through the process of electrolysis theoretically requires 9 L of water per kg of hydrogen based on the stoichiometric values.

- $2H_2O \longrightarrow O_2 + 4H^+ + 4e^-$
- $4H^+ + 4e^- \longrightarrow 2H_2$
- Overall: $2H_2O \longrightarrow O_2 + 2H_2$

However, most commercial electrolysis units on the market today advertise that they require between 10 and 11 L of deionized water per kg of hydrogen produced.

How much hydrogen is needed to produce Ammonia?

To understand how much hydrogen is needed to produce 1 kg of ammonia, we need to first understand the chemical reaction that produces ammonia. The equation for this reaction is:



This equation tells us that one molecule of nitrogen (N_2) reacts with three molecules of hydrogen (H_2) to form two molecules of ammonia (NH_3).

To convert this equation to a more practical quantity, we can use moles. A mole is a unit of measurement that represents a specific number of particles (atoms, molecules, or ions) of a substance. In this case, we can interpret the equation as follows: 1 mole of nitrogen reacts with 3 moles of hydrogen to produce 2 moles of ammonia.

The coefficients in the balanced equation tell us the relative number of moles of each substance involved in the reaction. So, if we know the mass of each substance involved in the reaction, we can use the molar masses to convert to moles, and then use the coefficients to calculate the amount of each substance needed to produce a certain amount of product.

According to the law of conservation of mass, the mass of reactants must equal the mass of products. To determine the mass of each substance involved in the reaction, we need to know the molar mass of each substance. The molar mass is the mass of one mole of a substance and is expressed in grams per mole (g/mol), and we can find this information in the periodic table.

The molar mass of nitrogen (N_2) is 28.02 g/mol (2×14.01). Similarly, the molar mass of hydrogen (H_2) is 2.02 g/mol (2×1.01), and the molar mass of ammonia (NH_3) is 17.03 g/mol ($1 \times 14.01 + 3 \times 1.01$).

Note: remember to be consistent with the number of decimals used, if you are taking only two for N and O, you can approximate H's.

Using these values, we can calculate the mass of each substance involved in the reaction:

$$\begin{array}{rcl} 1 \text{ mole of nitrogen gas (N}_2\text{)} & = & 28.02 \text{ g} \\ 3 \text{ moles of hydrogen gas (3 H}_2\text{)} & = & 6.06 \text{ g} \\ 2 \text{ moles of ammonia (2 NH}_3\text{)} & = & 28.02 + 6.06 = 34.08 \text{ g} \end{array}$$

We know that as for every 2 moles of ammonia produced, 3 moles of hydrogen are required. We also know the mass of the reactants and the product. Therefore, to produce 1 kg (1000 g) of ammonia, the amount of hydrogen we need is:

$$\frac{6.06 \text{ g} \times 1000 \text{ g}}{34.08 \text{ g}} = 177.82 \text{ g} \quad (3)$$

In other words, we need approximately 0.177 kg of hydrogen to produce 1 kg of ammonia.

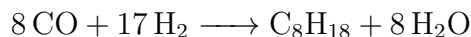
Since we need 9 L of water per kg of hydrogen (1 L = 1 kg water)

$$9 \cdot 0.17782 = 1.6$$

We need approximately 1.6 kg of water to produce 1 kg of ammonia.

It's important to note that this calculation assumes perfect efficiency in the electrolysis process and does not take into account any losses or inefficiencies that may occur in practice.

CHALLENGE: You can now do your own calculations for the use of hydrogen to create 1 kg of jet fuel (kerosene).



Calculation Question

A synfuel plant uses the Fischer-Tropsch process to convert hydrogen (H_2) and carbon monoxide (CO) into octane (C_8H_{18}). The plant operates with the following parameters:

- Hydrogen production rate: 1000 kg/day
- Carbon monoxide feed rate: 3000 kg/day
- Fischer-Tropsch reaction efficiency: 85%
- Molar mass of H_2 : 2 g/mol
- Molar mass of CO : 28 g/mol
- Molar mass of octane (C_8H_{18}): 114 g/mol
- Cost of hydrogen: \$10 NZD/kg
- Cost of carbon monoxide: \$5 NZD/kg
- Selling price of octane: \$50 NZD/kg

Calculate the daily production of octane (in kg) and the net profit assuming complete conversion of reactants and given efficiency.

Solution

1. Convert kg/day to g/day:

$$\text{H}_2 : 1000 \text{ kg/day} \times 1000 \text{ g/kg} = 1,000,000 \text{ g/day}$$

$$\text{CO} : 3000 \text{ kg/day} \times 1000 \text{ g/kg} = 3,000,000 \text{ g/day}$$

2. Calculate moles:

$$\text{H}_2 : \frac{1,000,000 \text{ g}}{2 \text{ g/mol}} = 500,000 \text{ mol}$$

$$\text{CO} : \frac{3,000,000 \text{ g}}{28 \text{ g/mol}} = 107,143 \text{ mol}$$

3. Determine limiting reactant:

CO is the limiting reactant: 107,143 mol

4. Calculate moles of octane produced:

$$\text{Synthetic fuel: } \frac{107,143 \text{ mol}}{8} = 13,393 \text{ mol}$$

5. Convert moles to grams:

$$\text{Synthetic fuel: } 13,393 \text{ mol} \times 114 \text{ g/mol} = 1,526,802 \text{ g}$$

6. Apply efficiency:

$$\text{Effective grams: } 1,526,802 \text{ g} \times 0.85 = 1,297,782 \text{ g}$$

7. Convert grams to kilograms:

$$\text{Synthetic fuel: } \frac{1,297,782 \text{ g}}{1000} = 1,297.78 \text{ kg}$$

8. Calculate costs:

$$\text{Cost of hydrogen: } 1000 \text{ kg} \times 10 \text{ NZD/kg} = 10,000 \text{ NZD}$$

$$\text{Cost of carbon monoxide: } 3000 \text{ kg} \times 5 \text{ NZD/kg} = 15,000 \text{ NZD}$$

9. Calculate revenue:

$$\text{Revenue from octane: } 1,297.78 \text{ kg} \times 50 \text{ NZD/kg} = 64,889 \text{ NZD}$$

10. Calculate net profit:

$$\text{Net profit: } 64,889 \text{ NZD} - (10,000 \text{ NZD} + 15,000 \text{ NZD}) = 39,889 \text{ NZD}$$

Answer

The daily production of octane is 1,297.78 kg, and the net profit is \$39,889 NZD.